📌 Project Title

Campus Innovation & Collaboration Portal

A web-based Java application where students can submit project ideas, collaborate with teammates, vote on ideas, and get faculty approvals.

🧱 COMPLETE TECH STACK

1️⃣ Programming Language

Java (JDK 17 or later)

Used for:

- Backend logic
- Controllers
- Services
- Data models
- Security configuration

2️⃣ Backend Framework

Spring Boot

Purpose:

- Build Java web applications
- Handle routing and API endpoints
- Manage dependencies
- Provide embedded server

Key features used:

- Spring MVC
- Spring Boot Auto Configuration
- Embedded Tomcat server

3️⃣ Security Framework

Spring Security

Used for:

- Login authentication

- Password encryption (BCrypt)
- Role-based access
- Session management
- Route protection

Roles:

ROLE_STUDENT

ROLE_FACULTY

ROLE_ADMIN

4️⃣ Database

MySQL

Stores:

- Users
- Projects
- Votes
- Team members
- Milestones

5️⃣ ORM Layer

Spring Data JPA + Hibernate

Purpose:

- Convert Java objects → database records
- Automatically generate SQL queries
- Manage entity relationships

Example mapping:

User.java → users table

Project.java → projects table

6️⃣ Frontend Template Engine

Thymeleaf

Used to render dynamic HTML pages.

Example:

<h1 th:text="${project.title}"></h1>

Allows backend data to be displayed in UI.

7️⃣ Frontend Technologies

HTML5

Structure of pages.

CSS3

Styling.

Bootstrap 5

Responsive layout.

Used for:

- Navbar
- Cards
- Grid system
- Buttons
- Forms
- Modals

8️⃣ Animation Libraries

AOS (Animate On Scroll)

Used for:

- Scroll animations
- Fade effects
- Smooth UI transitions

Example:

data-aos="fade-up"

9️⃣ Charts

Chart.js

Used in dashboard for:

- Project statistics
- Voting distribution
- Activity graphs

🔟 Build Tool

Maven

Manages:

- dependencies
- build process
- packaging

Command:

mvn clean package

Generates deployable file:

project.jar

1️⃣1️⃣ Version Control

Git + GitHub

Used for:

- collaborative development
- version tracking
- code merging

1️⃣2️⃣ Deployment Platform

Recommended:

Render

Supports Java applications easily.

Alternative:

- Railway
- AWS
- DigitalOcean

🏗 SYSTEM ARCHITECTURE

We will follow **Layered Architecture**.

Frontend (HTML + Bootstrap + JS + Animations)

↓

Thymeleaf Templates

↓

Controller Layer (Spring MVC)

↓

Service Layer (Business Logic)

↓

Repository Layer (Spring Data JPA)

↓

Hibernate ORM

↓

MySQL Database

📁 PROJECT STRUCTURE

```
campus-innovation-portal
|
├── src/main/java/com/campusportal
|
|   ├── controller
|   |   AuthController.java
|   |   ProjectController.java
|   |   VoteController.java
|   |   TeamController.java
|   |   FacultyController.java
|
|   ├── service
|   |   UserService.java
|   |   ProjectService.java
```

```
|   |    VoteService.java
|
|   ├── repository
|   |    UserRepository.java
|   |    ProjectRepository.java
|   |    VoteRepository.java
|
|   ├── model
|   |    User.java
|   |    Project.java
|   |    Vote.java
|   |    TeamMember.java
|
|   └── config
|        SecurityConfig.java
|
├── src/main/resources
|
|   ├── templates
|   |    login.html
|   |    register.html
|   |    dashboard.html
|   |    submit-project.html
|   |    project-list.html
|
|   ├── static
|   |    css
|   |    js
|   |    images
|
|   └── application.properties
```

|

└── pom.xml

🗄 **DATABASE DESIGN**

users

id

name

email

password

role

department

year

skills

---

projects

id

title

description

domain

status

created_by

vote_count

progress

---

team_members

id

project_id

user_id

status

---

votes

id

project_id

user_id

---

milestones

id

project_id

description

date

⚙️ IMPLEMENTATION WORKFLOW

1️⃣ User Registration

User fills form

↓

POST /auth/register

↓

AuthController receives request

↓

UserService validates data

↓

Password encrypted with BCrypt

↓

UserRepository saves user

↓

User stored in MySQL

2️⃣ User Login

User enters credentials

↓

Spring Security intercepts

↓

UserRepository fetches user

↓

Password verified

↓

Session created

↓

Redirect to dashboard

3️⃣ Submit Project

Student fills project form

↓

POST /projects/create

↓

ProjectController receives data

↓

ProjectService processes request

↓

ProjectRepository saves project

↓

Status = PENDING

4️⃣ Project Voting

Student clicks vote

↓

VoteController receives request

↓

VoteService checks duplicate vote

↓

VoteRepository stores vote

↓

Project vote_count updated

5️⃣ Join Project Team

Student clicks Join Project

↓

TeamController creates request

↓

TeamMember status = PENDING

↓

Project owner approves

↓

Status = ACCEPTED

6️⃣ Faculty Approval

Faculty views pending projects

↓

Approve or Reject

↓

FacultyController updates project status

↓

Status = APPROVED / REJECTED

7️⃣ Progress Tracking

Students update:

milestones

completion percentage

Data saved in **milestones table**.

Dashboard shows **progress bars**.

📊 Dashboard Analytics

Admin dashboard displays:

- total projects

- approved projects
- active students
- voting statistics

Using **Chart.js**.

🌐 DEPLOYMENT WORKFLOW

Complete development

↓

Push project to GitHub

↓

Connect repository to Render

↓

Render builds using Maven

↓

Spring Boot application deployed

↓

Public URL generated

Example:

https://campus-innovation-portal.onrender.com

🧠 FINAL SUMMARY

This project uses:

Language:

Java

Backend:

Spring Boot

Spring Security

Database:

MySQL

ORM:

JPA + Hibernate

Frontend:

HTML

CSS

Bootstrap

Thymeleaf

Animations:

AOS

Charts:

Chart.js

Build Tool:

Maven

Version Control:

GitHub

Deployment:

Render

The final system will be a **fully deployable Java web application with modern UI and collaborative features**.
